After 1967, the Soviet Union resupplied Egypt with MiGs and SAMs, while the United States became Israel's arms supplier, providing first McDonnell Douglas A-4s and then Phantoms. It was the start of a long period in which the Middle East would periodically provide combat trials for the latest Cold War aircraft and air-defense systems.

Electronic warfare

During the Six-Day War, guns and iron bombs had predominated over missiles, and electronic warfare had played only a limited role. But by 1973, when Israeli and Arab forces met again in full-scale conflict in the Yom Kippur war, missiles and the electronic countermeasures (ECM) to them were at the heart of the fighting. Soviet-supplied SAMs and radar-guided antiaircraft guns initially inflicted heavy losses on Israeli airmen hastily thrown into strike against Egyptian troops who had crossed the Suez Canal and Syrian forces advancing on the Golan Heights. The Israelis lost 40 aircraft in the first few hours of the war – about a tenth of their total air force.

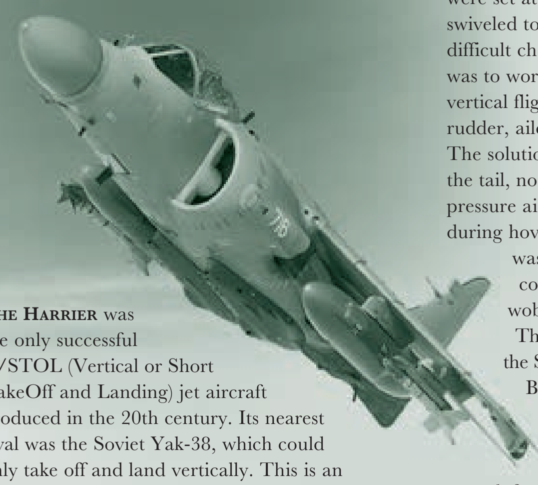
Although the quality of the Israeli pilots ensured that they maintained overwhelming supremacy in air-to-air combat, it was only when the Americans provided them with the latest ECM pods for their F-4s that they were able to overcome the gun-and-missile air-defense systems and once more assert total air superiority over Egypt and Syria. The lesson was clear: even the best-trained and motivated pilots would need the most up-to-date equipment if they were to survive on the evolving electronic battlefield.

Vertical takeoff and landing

One effect of the 1967 Israeli preemptive strike was to make air planners extremely nervous about airfields. The fact that the Egyptian airplanes had been lined up on the airfield waiting to be shot up was not any special folly, but largely standard practice in the world's air forces. There was a rush to reorganize airfields so that aircraft were dispersed and to create bomb-proof shelters to house them. Yet the dependence of aircraft on their concrete runways remained a weak point that an enemy could exploit. If runways were cratered, aircraft could not operate even if they survived.

Fears about the vulnerability of airbases created fresh interest in VTOL (Vertical TakeOff and Landing) jets, which could operate from any flat space, although some were intended as shipborne fighters. There had been various attempts at creating viable VTOL fixed-wing aircraft. In the mid-1950s, a number of experimental "tail-sitters" were built – aircraft such as the Convair XFY-1 Pogo and Ryan X-13 Vertijet. Their upright

BAe Harrier

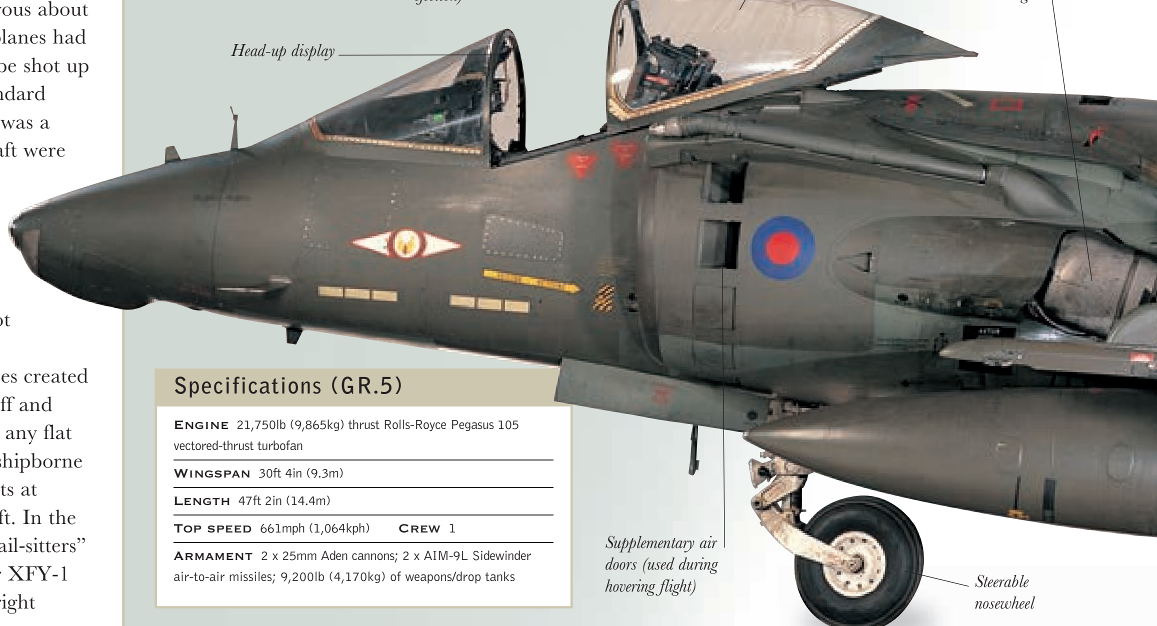


The Harrier was the only successful V/STOL (Vertical or Short TakeOff and Landing) jet aircraft produced in the 20th century. Its nearest rival was the Soviet Yak-38, which could only take off and land vertically. This is an important distinction because using vertical takeoff severely limits the amount of fuel and munitions that an aircraft can carry. Using a short takeoff enabled the Harrier "Jump Jet" to increase its range and weaponload. Therefore, the normal profile for a Harrier mission was STOVL – Short TakeOff and Vertical Landing.

The secret to the Harrier's success was its Pegasus vectored-thrust engine, which had four swiveling nozzles instead of the single fixed exhaust pipe found in other jets. The Yak-38 used lift engines that were a dead weight in forward flight. For vertical takeoff or landing the Harrier's nozzles were rotated to point downward. For short takeoff they were set at an angle. In normal flight they were swiveled to point backward. One of the most difficult challenges the Harrier's designers faced was to work out how to control the aircraft during vertical flight, when the usual control surfaces – the rudder, ailerons, and tailplane – were inoperative. The solution was to place pilot-operated valves in the tail, nose, and wingtips that release jets of high-pressure air to control the aircraft's pitch and roll during hovering flight. However, using this system was by no means easy – one Harrier pilot compared it to balancing on top of four wobbly bamboo poles.

The Royal Navy version of the Harrier, the Sea Harrier, became operational in 1980. Because the Harrier was able to take off from a 600ft (180m) flight deck (with the assistance of an inclined ramp), aircraft carriers originally considered only big enough for helicopters were able to operate a fixed-wing aircraft. Also, it was Marine pilots who discovered that the swiveling nozzles could be used to produce rapid deceleration and other unexpected jinks unavailable to conventional fighters.

In the 1990s British Aerospace and Boeing developed the Harrier II Plus as an attack and fighter aircraft. It was still in service with the US Marines in 2016, awaiting replacement by the V/STOL variant of the Lockheed-Martin F-35, which showed the continuing faith in the V/STOL concept.



Specifications (GR.5)

Engine	21,750lb (9,865kg) thrust Rolls-Royce Pegasus 105 vectored-thrust turbofan
Wingspan	30ft 4in (9.3m)
Length	47ft 2in (14.4m)
Top speed	661mph (1,064kph) **Crew** 1
Armament	2 x 25mm Aden cannons; 2 x AIM-9L Sidewinder air-to-air missiles; 9,200lb (4,170kg) of weapons/drop tanks